

Vemurafenib: the road to personalized medicine in melanoma.

PMID: 22384451 | Indexed in PubMed

Advanced melanoma has a poor prognosis due to its resistance to traditional chemotherapeutics, leading to the search for alternative treatment approaches. With the finding that approximately 50% of melanomas harbor an activating mutation in the serine/threonine-protein kinase B-raf gene (BRAF), inhibition of mutated B-raf represented an attractive and innovative focus for the development of novel targeted therapy potentially benefiting a large proportion of melanoma patients. Impressive response rates with an overall survival benefit in addition to minimal treatment-related toxicity in phase I-III clinical studies led to the FDA's approval of vemurafenib for patients with locally advanced/unresectable or metastatic BRAFV600E-mutated malignant melanoma in August 2011. While the majority of patients with BRAF-mutated disease show favorable treatment responses shortly after initiation of vemurafenib therapy, the median progression-free survival is 6 months, making the search for resistance mechanisms a high priority. While vemurafenib represents an excellent model for successful targeted anticancer therapy, long-term safety data are needed and rational combination with other agents will be critical to prevent or circumvent the development of resistance.

[The effect of solar ultraviolet radiation (UVR) on induction of skin cancers].

PMID: 27221301 | Indexed in PubMed

Ultraviolet radiation is a physical mutagenic and cancerogenic factor. About 95% of ultraviolet A (UVA) (320-400 nm) and 5% of UVB (280-320 nm) reach the Earth's surface. Melanin is a natural skin protective factor against UV radiation. Skin cancers associated with long-term exposure to UV radiation are: basal cell carcinoma (BCC), squamous cell carcinoma (SCC) and cutaneous malignant melanoma (CMM). The high risk of BCC development is related to acute and repeated exposure to UV causing sunburn. Molecular studies of BCC demonstrated disorders in sonic hedgehog (SHH) cell signaling regulation pathway, associated with the suppressor protein patched homolog 1 gene (PTCH1) mutations. The risk of the BCC development is related to the polymorphism of melanokortin-1 receptor gene (MC1R). Tumor P53 gene mutations observed in BCC cells has been classified as secondary genetic changes. In SCC cells UV-induced mutations were mostly related to P53 gene. Increased expression of cyclooxygenase-2 gene (COX-2) plays a significant role in the development of SCC. Other pathogenetic factors include intensification of the synthesis of pro-inflammatory cytokines (tumor necrosis factor α (TNF- α), interleukin-1 α (IL-1 α), IL-1 β and IL-6). Currently, the role of UVB has been recognized in the pathogenesis of CMM. In CMM cells the following gene mutations were noted: cyclin-dependent kinase inhibitor 2A (INK4A) (p16INK4A), cyclin-dependent kinase 4 (CDK4), Ras, phosphatase and tensin homolog deleted on chromosome 10 (PTEN) and proto-oncogene B-Raf (BRAF). The BRAF gene mutations were observed in ~50% of CMM cases. Mutations of P53 gene are not characteristic of CMM cells. Med Pr 2016;67(2):255-266.

Insights into trypanosomiasis transmission: Age, infection rates, and bloodmeal analysis of Glossina fuscipes fuscipes in N.W. Uganda.

PMID: 39480870 | Indexed in PubMed

Tsetse flies (*Glossina*) transmit species of *Trypanosoma* which cause human African trypanosomiasis (HAT) and animal African trypanosomiasis (AAT). Understanding the epidemiology of this disease and controlling the vector rationally requires analysis of the abundance, age structure, infection rates and feeding patterns of tsetse populations. We analysed a population of *G. fuscipes fuscipes* in the Koboko district of Uganda. Seasonal variation in the abundance of tsetse was assessed from the numbers of tsetse caught in pyramidal traps. The age structure of the population was assessed by dissecting female tsetse to estimate their ovarian categories. Classical and PCR-based methods were utilised to determine the presence of the three major pathogenic species of salivarian trypanosomes: *T. vivax*, *T. congolense* and *T. brucei* in a subset ($n = 2369$) of flies. Further, bloodmeal analysis was carried out using PCR to amplify and sequence a portion of the vertebrate *cytb* gene. The abundance and age structure of tsetse populations were relatively stable and a slight seasonal four-fold variation in abundance appeared to be correlated with rainfall. Analyses of age structure suggests a low natural daily mortality of 1.75% (1.62-1.88). Infection rates estimated were significantly greater (1.9-9.3 times) using the PCR-based method compared to the classical dissection-based method. Positive rates for *T. brucei* sl, *T. congolense* and *T. vivax* were 1.6% (1.32-2.24), 2.4% (1.83-3.11) and 2.0% (1.46-2.63), respectively by PCR. The majority of bloodmeals were identified as cattle (39%, 30.5-47.8) and human (37%, 28.4-45.6). The seasonally stable abundance, low mortality rate and high proportion of bloodmeals from humans may explain, in part, why this district has historically been a focus of sleeping sickness. Additionally, the high rates of cattle feeding indicate insecticide treated cattle may prove to be a useful vector control strategy in the area.

The Pathogenesis of African Trypanosomiasis.

PMID: 36055769 | Indexed in PubMed

African trypanosomes are bloodstream protozoan parasites that infect mammals including humans, where they cause sleeping sickness. Long-lasting infection is required to favor parasite transmission between hosts. Therefore, trypanosomes have developed strategies to continuously escape innate and adaptive responses of the immune system, while also preventing premature death of the host. The pathology linked to infection mainly results from inflammation and includes anemia and brain dysfunction in addition to loss of specificity and memory of the antibody response. The serum of humans contains an efficient trypanolytic factor, the membrane pore-forming protein apolipoprotein L1 (APOL1). In the two human-infective trypanosomes, specific parasite resistance factors inhibit APOL1 activity. In turn, many African individuals express APOL1 variants that counteract these resistance factors, enabling them to avoid sleeping sickness. However, these variants are associated with chronic kidney disease, particularly in the context of virus-induced inflammation such as coronavirus disease 2019. Vaccination perspectives are discussed.

The bloodstream form of *Trypanosoma brucei* displays non-canonical gluconeogenesis.

PMID: 38394337 | Indexed in PubMed

Trypanosoma brucei is a causative agent of the Human and Animal African Trypanosomiases. The mammalian stage parasites infect various tissues and organs including the bloodstream, central nervous system, skin, adipose tissue and lungs. They rely on ATP produced in glycolysis, consuming large amounts of glucose, which is readily available in the mammalian host. In addition to glucose, glycerol can

also be used as a source of carbon and ATP and as a substrate for gluconeogenesis. However, the physiological relevance of glycerol-fed gluconeogenesis for the mammalian-infective life cycle forms remains elusive. To demonstrate its (in)dispensability, first we must identify the enzyme(s) of the pathway. Loss of the canonical gluconeogenic enzyme, fructose-1,6-bisphosphatase, does not abolish the process hence at least one other enzyme must participate in gluconeogenesis in trypanosomes. Using a combination of CRISPR/Cas9 gene editing and RNA interference, we generated mutants for four enzymes potentially capable of contributing to gluconeogenesis: fructose-1,6-bisphosphatase, sedoheptulose-1,7-bisphosphatase, phosphofructokinase and transaldolase, alone or in various combinations. Metabolomic analyses revealed that flux through gluconeogenesis was maintained irrespective of which of these genes were lost. Our data render unlikely a previously hypothesised role of a reverse phosphofructokinase reaction in gluconeogenesis and preclude the participation of a novel biochemical pathway involving transaldolase in the process. The sustained metabolic flux in gluconeogenesis in our mutants, including a triple-null strain, indicates the presence of a unique enzyme participating in gluconeogenesis. Additionally, the data provide new insights into gluconeogenesis and the pentose phosphate pathway, and improve the current understanding of carbon metabolism of the mammalian-infective stages of *T. brucei*.

Prevalence of blood and skin trypanosomes in domestic and wild fauna from two sleeping sickness foci in Southern Cameroon.

PMID: 37498955 | Indexed in PubMed

Although studies on African Trypanosomiasis revealed a variety of trypanosome species in the blood of various animal taxa, animal reservoirs of *Trypanosoma brucei gambiense* and anatomical niches such as skin have been overlooked in most epidemiological settings. This study aims to update epidemiological data on trypanosome infections in animals from human African trypanosomiasis (HAT) foci of Cameroon. Blood and skin snips were collected from 291 domestic and wild animals. DNA was extracted from blood and skin snips and molecular approaches were used to identify different trypanosomes species. Immunohistochemical analyses were used to confirm trypanosome infections in skin snips. PCR revealed 137 animals (47.1%) with at least one trypanosome species in the blood and/or in the skin. Of these 137 animals, 90 (65.7%) and 32 (23.4%) had trypanosome infections respectively in the blood and skin. Fifteen (10.9%) animals had trypanosome infections in both blood and skin snip. Animals from the Campo HAT focus (55.0%) were significantly ($X^2 = 17.6$; $P < 0.0001$) more infected than those (29.7%) from Bipindi. Trypanosomes of the subgenus *Trypanozoon* were present in 27.8% of animals while *T. vivax*, *T. congolense* forest type and savannah type were detected in 16.5%, 10.3% and 1.4% of animals respectively. *Trypanosoma b. gambiense* infections were detected in the blood of 7.6% (22/291) of animals. No *T. b. gambiense* infection was detected in skin. This study highlights the presence of several trypanosome species in the blood and skin of various wild and domestic animals. Skin appeared as an anatomical reservoir for trypanosomes in animals. Despite methodological limitations, pigs, sheep, goats and wild animals were confirmed as potential reservoirs of *T. b. gambiense*. These animal reservoirs must be considered for the designing of control strategies that will lead to sustainable elimination of HAT.

Human African trypanosomiasis in endemic populations and travellers.

PMID: 21901632 | Indexed in PubMed

Human African trypanosomiasis (HAT) or sleeping sickness is caused by the protozoan parasites *Trypanosoma brucei* (T.b.) *gambiense* (West African form) and *T.b. rhodesiense* (East African form) that are transmitted by the bite of the tsetse fly, *Glossina* spp.. Whereas most patients in endemic populations are infected with *T.b. gambiense*, most tourists are infected with *T.b. rhodesiense*. In endemic populations, *T.b. gambiense* HAT is characterized by chronic and intermittent fever, headache, pruritus, and lymphadenopathy in the first stage and by sleep disturbances and neuro-psychiatric disorders in the second stage. Recent descriptions of the clinical presentation of *T.b. rhodesiense* in endemic populations show a high variability in different foci. The symptomatology of travellers is markedly different from the usual textbook descriptions of African HAT patients. The onset of both infections is almost invariably an acute and febrile disease. Diagnosis and treatment are difficult and rely mostly on old methods and drugs. However, new molecular diagnostic technologies are under development. A promising new drug combination is currently evaluated in a phase 3 b study and further new drugs are under evaluation.

Molecular epidemiological studies on animal trypanosomiasis in Ghana.

PMID: 23025330 | Indexed in PubMed

African trypanosomes are extracellular protozoan parasites that are transmitted between mammalian hosts by the bite of an infected tsetse fly. Human African Trypanosomiasis (HAT) or sleeping sickness is caused by *Trypanosoma brucei rhodesiense* or *T. brucei gambiense*, while African Animal Trypanosomiasis (AAT) is caused mainly by *T. vivax*, *T. congolense*, *T. simiae*, *T. evansi* and *T. brucei brucei*. Trypanosomiasis is of public health importance in humans and is also the major constraint for livestock productivity in sub-Saharan African countries. Scanty information exists about the trypanosomiasis status in Ghana especially regarding molecular epidemiology. Therefore, this study intended to apply molecular tools to identify and characterize trypanosomes in Ghana. A total of 219 tsetse flies, 248 pigs and 146 cattle blood samples were collected from Adidome and Koforidua regions in Ghana in 2010. Initial PCR assays were conducted using the internal transcribed spacer one (ITS1) of ribosomal DNA (rDNA) primers, which can detect most of the pathogenic trypanosome species and *T. vivax*-specific cathepsin L-like gene primers. In addition, species- or subgroup-specific PCRs were performed for *T. b. rhodesiense*, *T. b. gambiense*, *T. evansi* and three subgroups of *T. congolense*. The overall prevalence of trypanosomes were 17.4% (38/219), 57.5% (84/146) and 28.6% (71/248) in tsetse flies, cattle and pigs, respectively. *T. congolense* subgroup-specific PCR revealed that *T. congolense* Savannah (52.6%) and *T. congolense* Forest (66.0%) were the endemic subgroups in Ghana with 18.6% being mixed infections. *T. evansi* was detected in a single tsetse fly. Human infective trypanosomes were not detected in the tested samples. Our results showed that there is a high prevalence of parasites in both tsetse flies and livestock in the study areas in Ghana. This enhances the need to strengthen control policies and institute measures that help prevent the spread of the parasites.

Human African trypanosomiasis.

PMID: 19833383 | Indexed in PubMed

Human African trypanosomiasis (sleeping sickness) occurs in sub-Saharan Africa. It is caused by the protozoan parasite *Trypanosoma brucei*, transmitted by tsetse flies. Almost all cases are due to *Trypanosoma brucei gambiense*, which is indigenous to west and central Africa. Prevalence is strongly dependent on control measures, which are often neglected during periods of political instability, thus

leading to resurgence. With fewer than 12 000 cases of this disabling and fatal disease reported per year, trypanosomiasis belongs to the most neglected tropical diseases. The clinical presentation is complex, and diagnosis and treatment difficult. The available drugs are old, complicated to administer, and can cause severe adverse reactions. New diagnostic methods and safe and effective drugs are urgently needed. Vector control, to reduce the number of flies in existing foci, needs to be organised on a pan-African basis. WHO has stated that if national control programmes, international organisations, research institutes, and philanthropic partners engage in concerted action, elimination of this disease might even be possible.

The continuing problem of human African trypanosomiasis (sleeping sickness).

PMID: 18756506 | Indexed in PubMed

Human African trypanosomiasis, also known as sleeping sickness, is a neglected disease, and it continues to pose a major threat to 60 million people in 36 countries in sub-Saharan Africa. Transmitted by the bite of the tsetse fly, the disease is caused by protozoan parasites of the genus *Trypanosoma* and comes in two types: East African human African trypanosomiasis caused by *Trypanosoma brucei rhodesiense* and the West African form caused by *Trypanosoma brucei gambiense*. There is an early or hemolymphatic stage and a late or encephalitic stage, when the parasites cross the blood-brain barrier to invade the central nervous system. Two critical current issues are disease staging and drug therapy, especially for late-stage disease. Lumbar puncture to analyze cerebrospinal fluid will remain the only method of disease staging until reliable noninvasive methods are developed, but there is no widespread consensus as to what exactly defines biologically central nervous system disease or what specific cerebrospinal fluid findings should justify drug therapy for late-stage involvement. All four main drugs used for human African trypanosomiasis are toxic, and melarsoprol, the only drug that is effective for both types of central nervous system disease, is so toxic that it kills 5% of patients who receive it. Eflornithine, alone or combined with nifurtimox, is being used increasingly as first-line therapy for gambiense disease. There is a pressing need for an effective, safe oral drug for both stages of the disease, but this will require a significant increase in investment for new drug discovery from Western governments and the pharmaceutical industry.